

Credibility, Strategic Communication, and Information Design

EC 2024 Tutorial

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Important feature of communication: Involves multiple agents.

At bare minimum, need to have two:

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Raises the question: What prevents full information transmission?

CLASSIC FRICTION: DECEPTION

Classic answer: S often wants to lie to R.

Most of the classic literature asks: what prevents S from lying?

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Most of the classic literature asks: what prevents S from lying?

Specifically, how can S's communication become *credible*?

- Repeated interactions.
- Costly signaling.
- Verifiable disclosure.
- Mediation.
- ...

BAYESIAN PERSUASION: COMMUNICATION WITHOUT DECEPTION

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Key observation: S often benefits from *hiding* information.

- This is a strategic impediment unrelated to credibility.
- Similar observation made by Rayo and Segal (2010).

Starting point for an extensive literature on *information design*.

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BP provides a clear benchmark for evaluating role of credibility.

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Q: How do credibility constraints impact information transmission?

Note: Credibility no longer guaranteed to reduce informativeness!

DIFFERENT MODES OF ANALYSIS

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Some classic analysis of strategic communication relies on beliefs...

- Games of incomplete information, long cheap talk...

... but most do not.

TODAY: BELIEF-BASED APPROACH IN STRATEGIC COMMUNICATION

So, we have two ways to go:

- Apply “classic approach” to Bayesian Persuasion.
- Apply “belief-based approach” to partial credibility models.

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Today’s tutorial:

- **Part 1:** Guaranteed credibility vs. no credibility.
- **Part 2:** Communication with partial credibility.

THE PLAN FOR PART 1

1. Full commitment case.

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- Introduce/review belief-based approach.
- Famous concavification result.

2. No commitment case: state-independent S-preferences.

- “Cheap Talk with Transparent Motives” (Lipnowski & Ravid, 2020).
- Special case of cheap talk amenable to belief-based analysis.
- Securability $\implies$ Quasiconcavification.
- Application: the “Think Tank.”

The Full Commitment Case

BAYESIAN PERSUASION MODEL

$$\Theta, \quad \mu_0 \in \Delta\Theta, \quad A, \quad M, \quad u_R : A \times \Theta \rightarrow \mathbb{R}, \quad u_S : A \times \Theta \rightarrow \mathbb{R}.$$

Timing:

1. S commits to a Blackwell experiment $\xi : \Theta \rightarrow \Delta M$.
2. Nature draws $\theta \in \Theta$ and $m \in M$.
3. R sees m and takes action $a \in A$.

(for the tutorial, all spaces are finite, and $|M| \geq \max\{|A|, |\Theta|\}$)

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$$\pi(\theta|m) \int \xi(m|\tilde{\theta}) \mu_0(d\tilde{\theta}) = \xi(m|\theta) \mu_0(\theta),$$

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Note: **S-incentives** do not enter equilibrium.

THE INFORMATION DESIGN PROGRAM

The goal is to find S's favorite PBE:

$$\begin{aligned} v^* = \max_{(\xi, \pi, \alpha)} & \int u_S(a, \theta) \alpha(da|m) \xi(dm|\theta) \mu_0(d\theta) \\ \text{s.t. } & (\xi, \pi, \alpha) \text{ is a PBE.} \end{aligned}$$

TOWARDS A SOLUTION: THE VALUE CORRESPONDENCE

For a belief $\mu \in \Delta\Theta$, extend players payoffs using expectations:

$$u_S(a, \mu) = \int u_S(a, \theta) \mu(d\theta), \quad \text{and} \quad u_R(a, \mu) = \int u_R(a, \theta) \mu(d\theta).$$

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S's (interim) **value correspondence** is

$$V : \Delta\Theta \rightrightarrows \mathbb{R},$$
$$V(\mu) = \text{co } u_S \left(\arg \max_{a \in A} u_R(a, \mu), \mu \right).$$

REFORM SELECTION

Think tank (S) advises lawmaker (R) on a reform.

Lawmaker can choose one of n reforms or status quo.

Reforms' values: unknown to lawmaker.

Think tank can release report about each reform's value.

Report's content is flexible (can vary methods & presentation).

Think tank has a commitment power: can commit to *any* report.

REFORM SELECTION

Think tank (S) advises lawmaker (R): status quo and n reforms.

$$A = \{0, 1, \dots, n\}.$$

$$\Theta = [0, 1]^n.$$

$$u_R(a, \theta) = \begin{cases} \theta_i - c & : a = i \in \{1, \dots, n\}, \\ 0 & : a = 0. \end{cases}$$

$$u_S(a, \theta) = a.$$

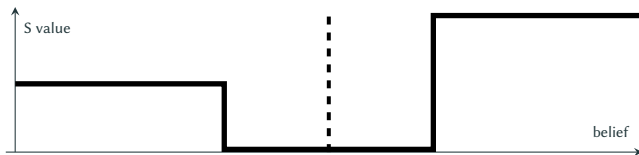
$$\mu_0 \in \Delta\Theta \text{ exchangeable.}$$

A TOY EXAMPLE

Simplest nontrivial special case:

$$\begin{aligned} n &= 2, \\ (\theta_1, \theta_2) &= \begin{cases} (1, 0) & : \text{w.p. } \frac{1}{2} \\ (0, 1) & : \text{w.p. } \frac{1}{2}, \end{cases} \end{aligned}$$

$$c = \frac{2}{3}.$$

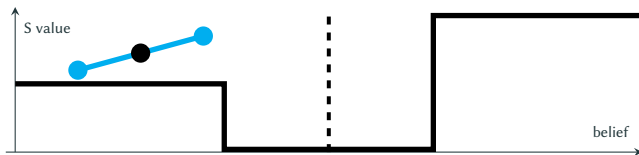


CHARACTERIZING EQUILIBRIUM OUTCOMES

In equilibrium, any message m induces:

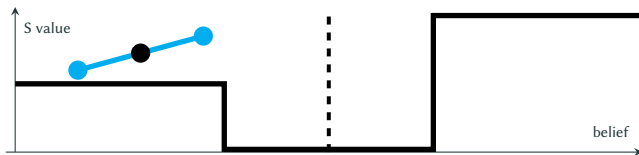
- An R *belief*, μ
- An R mixed action, which yields an S *interim payoff*, s .

Goal: characterize the PBE distributions of (*belief*, *S-payoff*) pairs.



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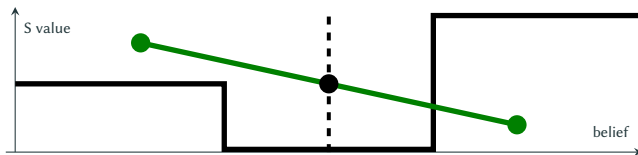
(i) **Bayes rule:** the $\Delta\Theta$ coordinate must average to the prior.



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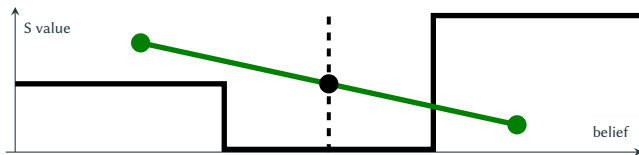
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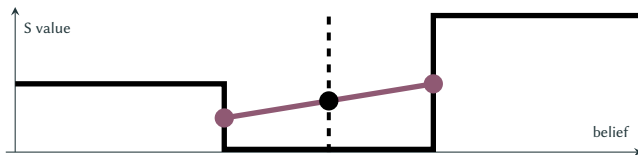
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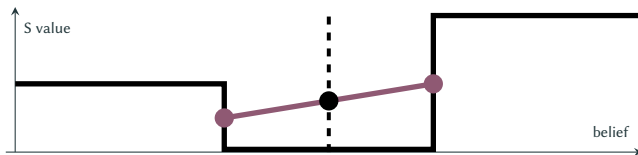
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$\mathbf{P} \in \Delta((\Delta\Theta) \times \mathbb{R})$ can arise in a PBE only if:

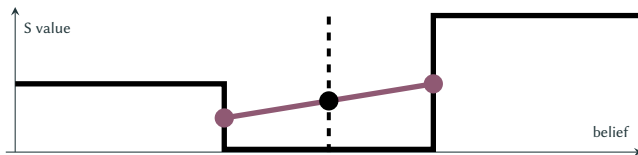
$$\int \mu \mathbf{P}(d(\mu, s)) = \mu_0, \quad \text{and} \quad \mathbf{P}(\text{graph } V) = 1.$$



CHARACTERIZING EQUILIBRIUM OUTCOMES

Lemma. $\mathbf{P} \in \Delta((\Delta\Theta) \times \mathbb{R})$ can arise in a PBE if and only if:

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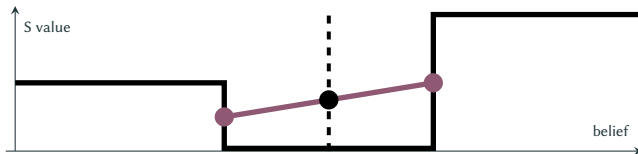


BELIEF-BASED PROGRAM

So, we can rephrase our problem:

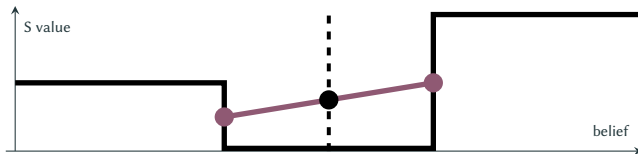
$$v^* = \max_{\mathbf{P} \in \Delta((\Delta\Theta) \times \mathbb{R})} \int s \mathbf{P}(d(\mu, s))$$

s.t. $\int \mu \mathbf{P}(d(\mu, s)) = \mu_0$, and $\mathbf{P}(\text{graph } V) = 1$.



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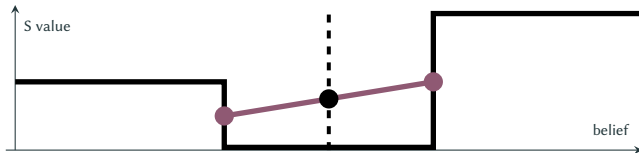


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For every μ , we always want to set

$$s = \max V(\mu) =: v(\mu) \quad (\text{“the value function”}).$$

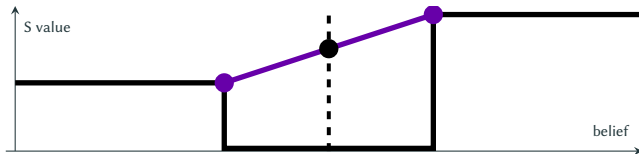


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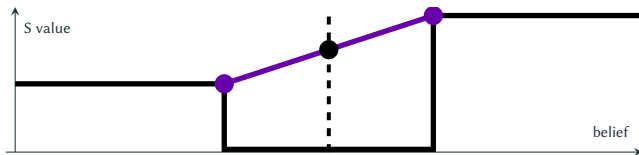
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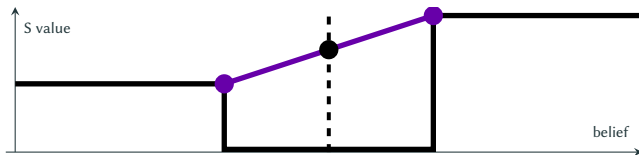
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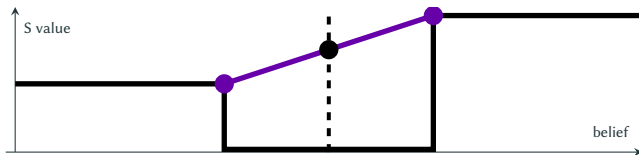
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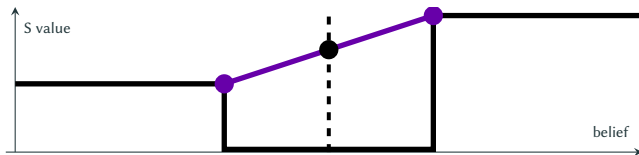
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- Computationally tractable (Dughmi and Xu, 2016).
- Can be solved geometrically via concavification.

CONCAVIFICATION

Take $\hat{v} : \Delta\Theta \rightarrow \mathbb{R}$ be the concave envelope of v .

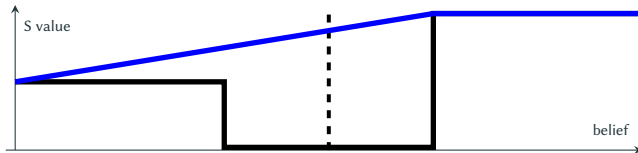
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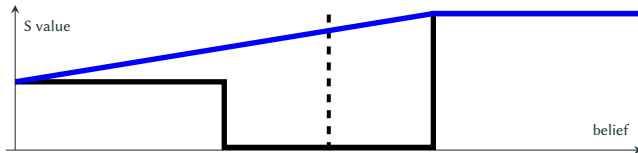


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Theorem. $v^* = \hat{v}(\mu_0)$.



The No Commitment Case

THE CHEAP TALK MODEL

What if S could not commit?

Version with least commitment power: cheap talk.

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Hence, the timing is now:

1. S sees $\theta \in \Theta$, and sends R a message m ,
2. R sees m , and takes action $a \in A$.

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A **perfect Bayesian equilibrium** (PBE) is

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Note: Different S-IC constraint for every θ .

These constraints make it hard to use belief-based approach.

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An exceptional special case: Transparent Motives,

$$u_S(a, \theta) = u_S(a).$$

- Politicians, sales-people, job applicants...

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An exceptional special case: **Transparent Motives**,

$$u_S(a, \theta) = u_S(a).$$

In this case, S-IC is just

$$\sigma \left(\arg \max_{m \in M} \int u_S(a) \alpha(da|m)|\theta \right) = 1 \quad \forall \theta \in \Theta$$

THE TRANSPARENT MOTIVES CASE

Note: Different S-IC constraint for every θ .

These constraints make it hard to use belief-based approach.

- possible, but hard—see Aumann and Hart (2003).

An exceptional special case: **Transparent Motives**,

$$u_S(a, \theta) = u_S(a).$$

In this case, S-IC is just

$$\int u_S(a) \alpha(da|m) = \max_{m \in M} \int u_S(a) \alpha(da|m) \quad \text{almost surely.}$$

INFORMATION AND EX-ANTE S-PAYOFFS

To use the belief-based approach, focus on two objects:

1. $p \in \Delta\Delta\Theta$: distribution over R's posterior belief.
2. $s \in \mathbb{R}$: S's *ex-ante* payoff.

INFORMATION AND EX-ANTE S-PAYOFFS

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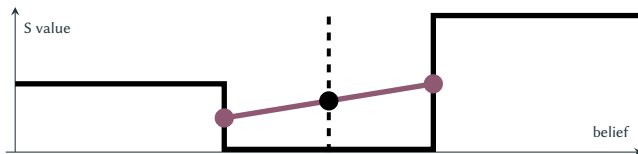
1. $p \in \Delta\Delta\Theta$: distribution over R's posterior belief.
2. $s \in \mathbb{R}$: S's *ex-ante* payoff.

Main Q: Which (p, s) can arise in a PBE?

CHARACTERIZING CHEAP TALK PBE OUTCOMES

Recall every PBE distributions of $(\text{belief}, S\text{-payoff})$ pairs has

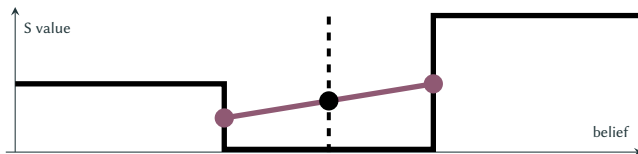
- (i) **Bayes rule**: the $\Delta\Theta$ coordinate must average to the prior.
- (ii) **R optimality**: we must live on the graph of V .



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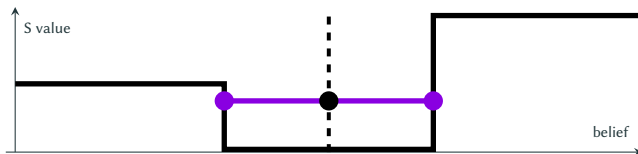
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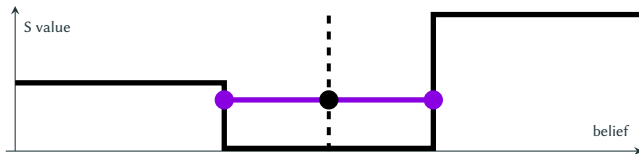


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Notice: (iii) $\implies$ S's *ex-ante* payoff equals her *interim* payoff!



CHARACTERIZING CHEAP TALK PBE OUTCOMES

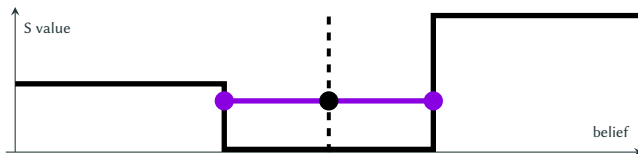
Lemma.

(p, s) can arise in a PBE if and only if

(i) $\int \mu \, p(d\mu) = \mu_0,$

(ii) $s \in \bigcap_{\mu \in \text{supp}(p)} V(\mu).$

Essentially follows from Aumann & Hart (2003).



SECURABILITY

Take

$$\mathcal{P}(\mu_0) = \left\{ p \in \Delta\Delta\Theta : \int \mu p(d\mu) = \mu_0 \right\}.$$



SECURABILITY

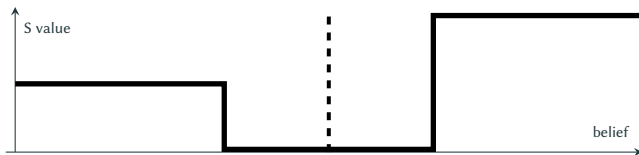
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Definition. Say $p \in \mathcal{P}(\mu_0)$ **secures** value $s \in \mathbb{R}$ if

$$p\{\mu : v(\mu) \geq s\} = 1.$$

Say s is **securable** if there is some $p \in \mathcal{P}(\mu_0)$ that secures s .



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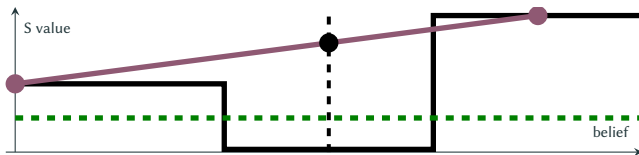
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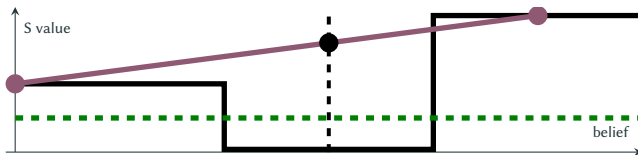
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SECURABILITY

Theorem (Lipnowski and Ravid, 2020). Let $s \geq v(\mu_0)$. Then,

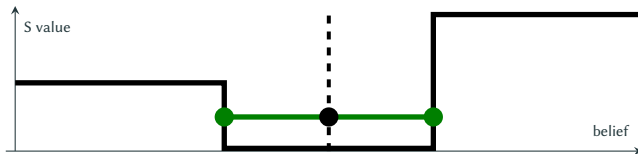
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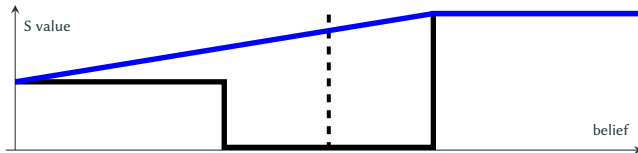


Full vs. No Commitment

COMMITMENT'S VALUE: GEOMETRIC PERSPECTIVE

Kamenica & Gentzkow (2011):

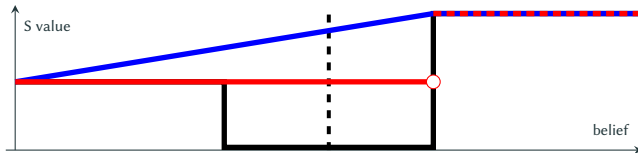
The highest S payoff attainable with S commitment is the **concave envelope** of v , evaluated at the prior.



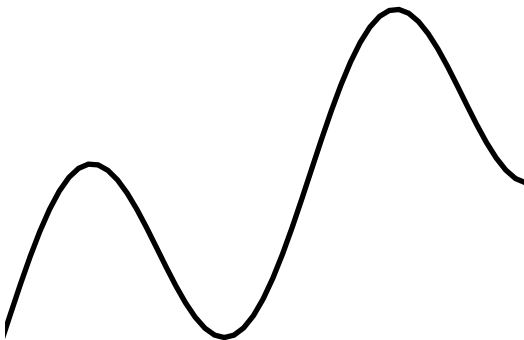
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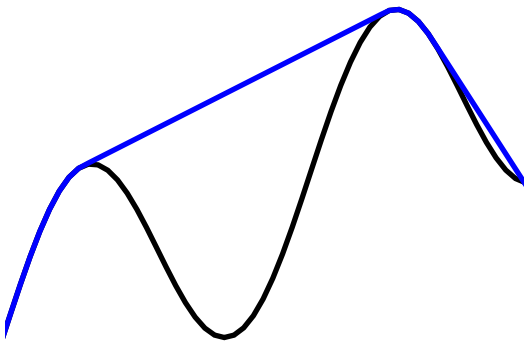
The highest S payoff attainable under cheap talk is the *quasiconcave envelope of v* , evaluated at the prior.



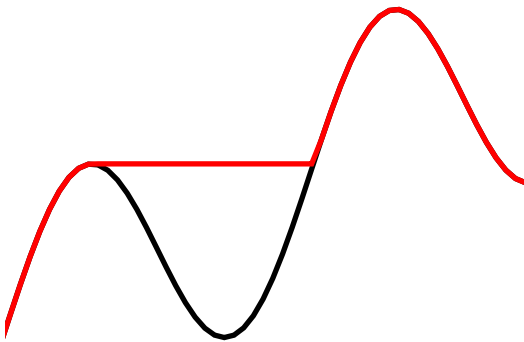
CONCAVE VS. QUASICONCAVE ENVELOPE



CONCAVE VS. QUASICONCAVE ENVELOPE



CONCAVE VS. QUASICONCAVE ENVELOPE



COMMITMENT IS USUALLY VALUABLE

Corollary. For Lebesgue-a.e. $\mu_0 \in \Delta\Theta$, one of the following holds:

(i) Cheap talk yields S her first-best value,

$$\bar{v}(\mu_0) = \max v(\Delta\Theta).$$

(ii) Commitment is valuable,

$$\bar{v}(\mu_0) < \hat{v}(\mu_0).$$

The Full Think Tank Example

REFORM SELECTION

Think tank (S) advises lawmaker (R) on a reform.

Lawmaker can choose one of n proposals or status quo.

Proposal's values: unknown to lawmaker.

Think tank has expertise: knows each proposal's value.

Also has **transparent motives**: strict preference over proposals.

REFORM SELECTION

Think tank (S) advises lawmaker (R): status quo and n proposals.

$$A = \{0, 1, \dots, n\}.$$

$$\Theta = [0, 1]^n.$$

$$u_R(a, \theta) = \begin{cases} \theta_i - c & : a = i \in \{1, \dots, n\}, \\ 0 & : a = 0. \end{cases}$$

$$u_S(a) = a \text{ (or something else increasing).}$$

$$\mu_0 \in \Delta\Theta \text{ exchangeable without ties.}$$

REFORM SELECTION

Claim: The following are equivalent.

- (i) $k \in \{1, \dots, n\}$ is an equilibrium S value.
- (ii) $\mathbb{E} [\max_{i \in \{k, \dots, n\}} \theta_i - c] \geq 0$.
- (iii) S can secure k by revealing the random variable

$$\mathbf{i}_k := \arg \max_{i \in \{k, \dots, n\}} \theta_i.$$

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Proof:

- (i) $\implies$ (ii): Lawmaker ex-ante incentives.
- (ii) $\implies$ (iii): Exchangeability.
- (iii) $\implies$ (i): By Securability.

REFORM SELECTION

Let $k^* := \max \{k \in \{1, \dots, n\} : \mathbb{E} [\max_{i \in \{k, \dots, n\}} \theta_i - c] \geq 0\}$.

The claim implies that

The S-preferred equilibrium value is k^* .

(If no such k^* exists, best is babbling.)

REFORM SELECTION

The following is an S favorite equilibrium.

- S names reform $\mathbf{i} \in \{k^*, \dots, n\}$ given by:

$$\begin{cases} \mathbf{i}_{k^*} = \arg \max_{i \in \{k^*, \dots, n\}} \theta_i & : \text{w.p. } 1 - \varepsilon, \\ \mathbf{i} \sim \text{uniform}\{k^*, \dots, n\} & : \text{w.p. } \varepsilon. \end{cases}$$

where $\varepsilon \in [0, 1)$ satisfies $(1 - \varepsilon)\mathbb{E} [\max_{i \in \{k, \dots, n\}} \theta_i] + \varepsilon\mathbb{E} [\theta_j] = c$.

- R indifferent between i and outside option. Chooses:

$$\begin{cases} i & : \text{w.p. } \frac{k^*}{i} \\ 0 & : \text{w.p. } 1 - \frac{k^*}{i}. \end{cases}$$

REFORM SELECTION

If $(\theta_i)_i$ are i.i.d. uniform, then:

$$\mathbb{E} \left[\max_{i \in \{k, \dots, n\}} \theta_i \right] = \frac{1+n-k}{2+n-k}.$$

Hence, solution (k^*, ε) is given by (if $c > \frac{1}{2}$):

$$k^* = \left\lfloor n - \frac{2c-1}{1-c} \right\rfloor,$$

$$\varepsilon = 2 \left[(1-c) - \frac{2c-1}{n-k^*} \right].$$

BOUNDING COMMITMENT'S VALUE

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Or, more generally,

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Note: solving the commitment case is difficult.

Securability and Computational Tractability

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- Bayesian Persuasion value is computationally tractable.

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Fun fact: can reduce (iii) to first paper using securability!

Concluding Remarks for Part 1

CONCLUDING PART 1 OF TUTORIAL

Bayesian Persuasion (Kamenica & Gentzkow, 2011):

- Asks: what information does S want to provide?
- A benchmark for studying credibility's role in comm.
- Model is very tractable, especially via the *belief-based approach*.
- Highest S value given by **concavification**.

Cheap Talk with Transparent Motives (Lipnowski & Ravid, 2020):

- A special case of cheap talk amenable to belief-based analysis.
- Can characterize S 's equilibrium values via **securability**.
- Highest S value given by **quasiconcavification**.
- Securability also useful for solving applications.